INCH-POUND

MIL-C-22442/35A
30 September 1991
SUPERSEDING
MIL-C-22442/35
14 November 1984

MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, RADIO FREQUENCY (AIRCRAFT AUDIO), CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for audio frequency cord assemblies with braided and molded elastomer jackets.

1.2 Military part number.

Molded Elastomer Jacket Coil Cord M22442/35-1.

Molded Elastomer Jacket Straight M22442/35-2.

Braided Jacket M22442/35-3 (INACTIVE FOR NEW DESIGN FOR NAVY USE).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-572	Cords, Yarns and Nonfilaments Organic, Synthetic Fiber
MIL-I-3930	Insulating and Jacketing Compounds, Electrical (for Cables, Cords and Wires)
MIL-C-55668	Cord, Electrical; Audio Subminiature (Retractile and Straight)

STANDARDS

MILITARY

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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STANDARDS (continued)

MIL-STD-454	Standard General Requirements for Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards) the text of this specification sheet takes precedence. Nothing in this specification sheet, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Fungus-resistant materials. Cords shall be made of fungus-resistant materials as specified in Requirement 4 of MIL-STD-454.

3.2 Design and construction.

3.2.1 Conductors. The nominal area of the conductors shall be 320 circular mils and the minimum 310 circular mils. Each conductor shall consist of 37 bunch-stranded AWG #40 (0.003145" diameter) tinned cadmium-bronze wires. The strands shall be twisted to have a left hand lay of .50 inch nominal. The individual strands, before bunching and tinning, shall have a minimum tensile strength of 90,000 pound-force per square inch and have a conductivity grade of at least 80 percent.

3.2.1.1 Tensile strength. The tensile strength of the conductor strands shall not decrease more than 10 percent after tinning and insulating when tested in accordance with 4.1.

3.2.2 Insulation. The insulation shall be melt-extruded of fluorinated ethylene propylene FEP having a minimum wall thickness of .006 inches.

3.2.3 Color code. The conductor insulation for the M22442/35-1 and M22442/35-2 cord assembly shall be colored white, black, red, green, black and white in that order, arranged in accordance with figure 1. The insulation for the M22442/35-3 cord assembly shall be colored red, green, white, orange and black, arranged in accordance with figure 2.

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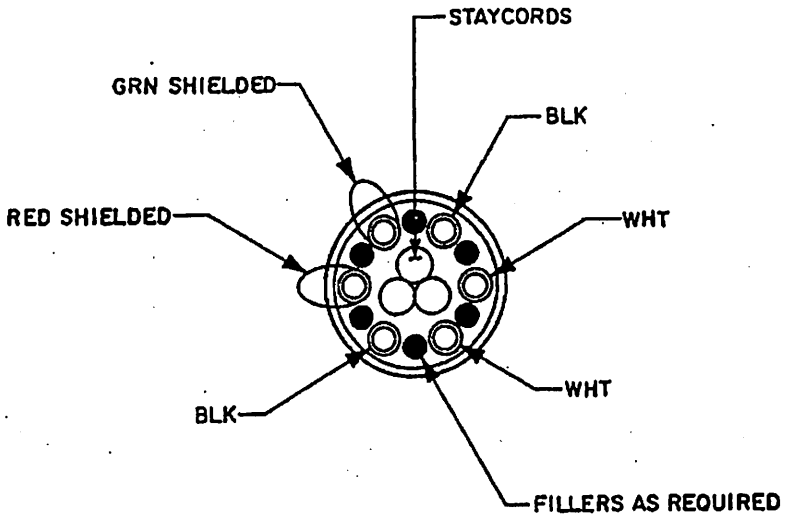


FIGURE 1. Cable construction P/N M22442/35-1 and P/N M22442/35-2.

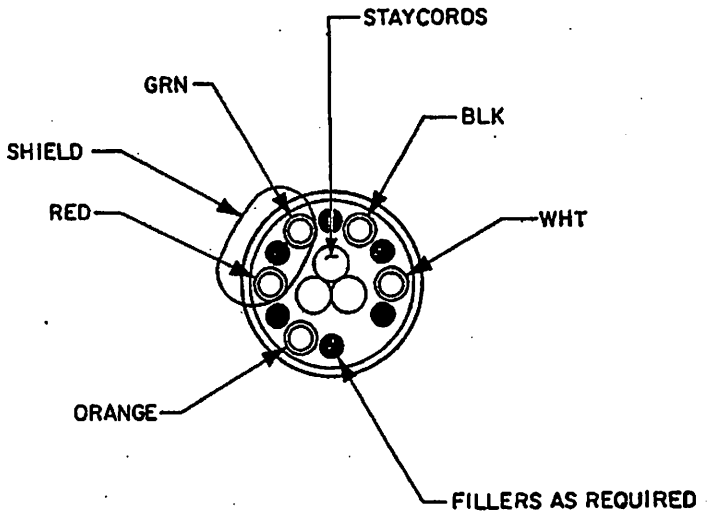


FIGURE 2. Cable construction P/N M22442/35-3.

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3.2.4 Stay cords. Stay cords shall consist of synthetic textile yarn conforming to Type P or Type PAA of MIL-C-572. The stay cords shall each have a minimum breaking strength of 20 pounds. Stay cords shall not abrade or otherwise damage the insulation of the conductors when the cord is bent or stretched.

3.2.5 Jacketing materials.

3.2.5.1 Braided. For M22442/35-3, the cable shall be covered with a braided jacket composed of black nylon, 210 denier, 3 ends, 24 carrier, approximately 30 picks per inch.

3.2.5.2 Thermoset elastomer. The M22442/35-1 or -2 cable shall have a jacket molded with a Type JS-L (SBR), JH (EPDM), JN or JNL compound per MIL-I-3930 or a SBR natural blend compound conforming to the performance requirements of MIL-C-55668. The jacket is to be applied in a continuous extrusion process or molded process directly onto the cable core. The jacket O.D. shall be in the range of .168 to .180 diameter with a jacket wall thickness of .016 to .025.

3.2.6 Shield. The shielding will be wrapped shielding, 85-95 percent coverage, using individual strands of size 36 AWG, nickel plated soft drawn copper strands.

3.2.7 Cable (cord) construction. Insulated conductors shall have a one-inch nominal right hand lay.

3.3 Performance.

3.3.1 Temperature. The cord shall be flexible and resilient throughout the temperature range of -55°C to +85° and shall not show evidence of cracking or other damage as a result of exposure to the temperature range specified.

3.3.2 Flex life. The cord assembly shall have a mean flex life of 25,000 flexes without showing evidence of:

a. Damage to the outside jacket.

b. Damage to the insulation of the individual conductors.

c. Electrical discontinuity.

3.3.3 Electrical characteristics.

3.3.3.1 Dielectric withstanding voltage. There shall be no evidence of breakdown when the cords are tested as specified in 4.5. Leakage currents shall be recorded and shall not vary more than 30 percent from initial values.

3.3.3.2 Insulation resistance. When tested as specified in 4.6, the insulation resistance shall not be less than 100 megohms.

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3.3.3.3 Electrostatic and electromagnetic isolation. The cord shall meet the electrostatic and electromagnetic isolation requirements listed in table I and when tested in accordance with figure 3 and 4.7.

TABLE I. Isolation requirements.

Frequency Hz	Electrostatic isolation (minimum decibels)	Electromagnetic isolation (minimum decibels)
20,000	106	75
10,000	112	81
5,000	118	87

3.3.4 Continuity. Each conductor in the cord shall be continuous when tested as specified in 4.8 and shall not exceed 4 millivolts drop above that of an equivalent length of the same conductor with a test current of 1.6 amperes (dc).

4. QUALITY ASSURANCE PROVISIONS

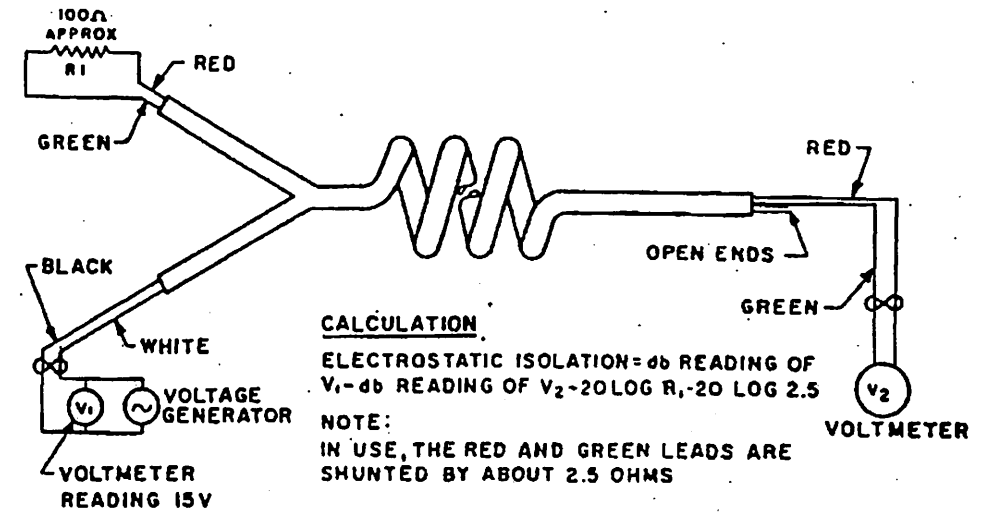
4.1 Tensile strength. The wire shall be tested to meet the requirements of 3.2.1.1. One strand selected at random from a conductor shall be tested.

4.2 Temperature. The unterminated cord shall be placed within a test chamber wherein a temperature of 85°C (185°F) is maintained for a period of four hours. The cord shall be wrapped about a mandrel while at this temperature without damage to the cord jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cord diameter. The cord shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cord shall be inspected for damage or deformation and then subjected to a test chamber wherein a temperature of -55°C (-67°F) is maintained. After 20 hours of storage at -55°C, the cord shall be wrapped about the mandrel while at this temperature, in the same manner as it was at +85°C. The cord shall be examined for cracks or any other damage to the jacket or conductor insulation. The cord shall meet the requirements of 3.3.1.

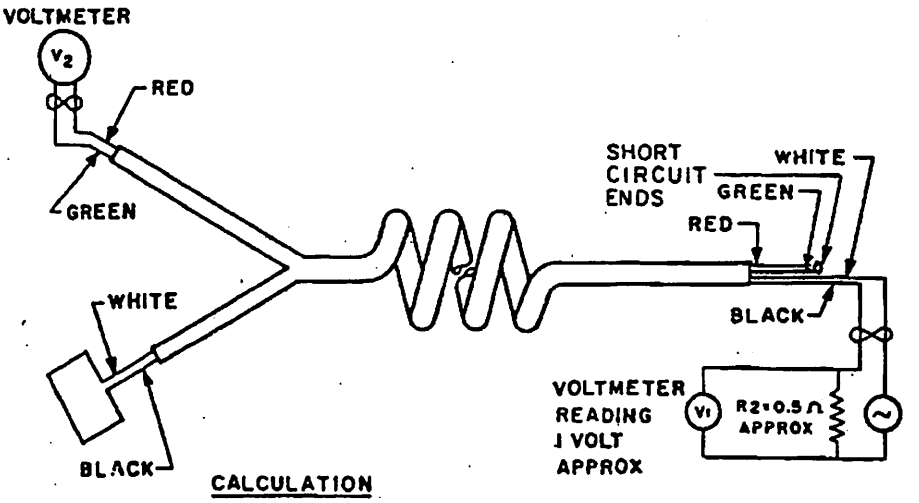
4.3 Flex life. Five samples of the unterminated cord shall be clamped and suspended through holes in a metal bar having a rectangular cross section of .50 inch by .625 inch as described below:

a. The clamping point shall be immediately above the metal bar; the cord clamp shall rest on top of the metal bar.

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ELECTROSTATIC ISOLATION TEST



ELECTROMAGNETIC ISOLATION TEST

NOTE: Test setup applies to straight cords also.

FIGURE 3. Test set-up.

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b. The cord shall be clamped in such a manner that it will not turn in the hole during the flexing test.

c. The length of the through holes in the metal bar shall be .625 inch. The diameters of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cord samples.

d. The through holes in the metal bar shall incorporate a .125 inch radius at the end from which the cord emerges (bending point).

e. A three-pound weight shall be hung from the cord at a point 8 to 12 inches below the bar to assure that the cord is held in a vertical position.

f. A small low-current drain, a 6-volt lamp, and a suitable power source shall be connected in series to each conductor to indicate electrical continuity (see figure 4). The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the five samples of the cord before electrical discontinuity occurs shall be taken as the flexing life of the cord (see 3.3.2).

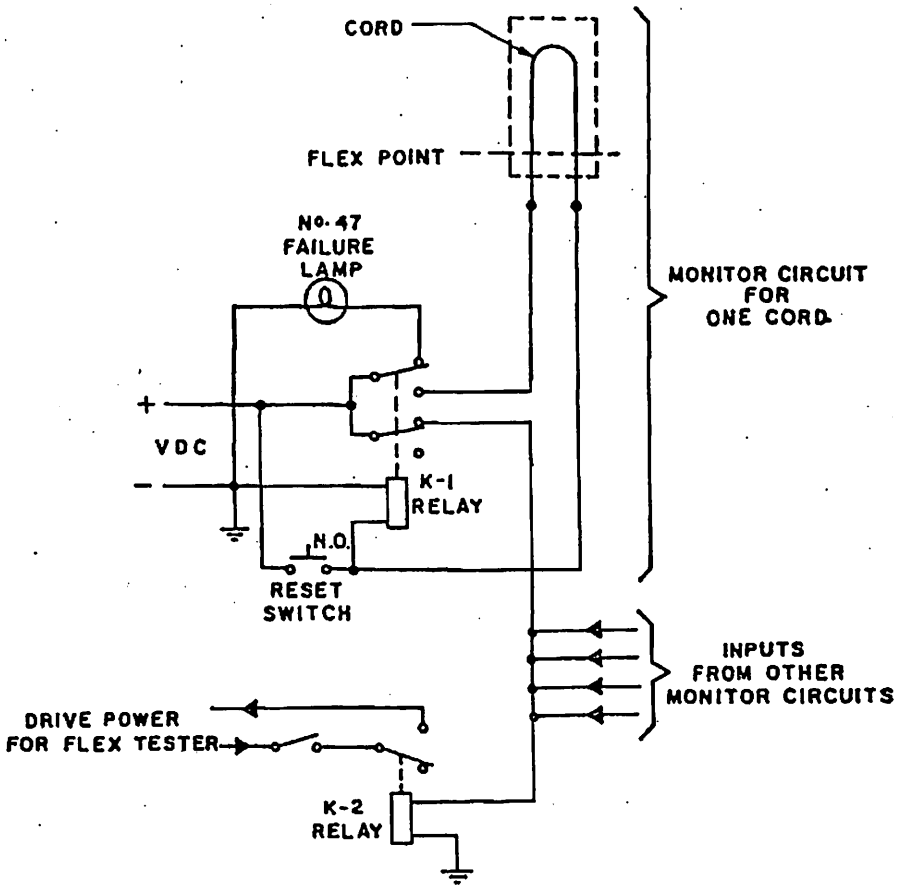
4.4 Dielectric withstanding voltage. Dielectric strength shall be performed in accordance with Method 301 of MIL-STD-202. A potential of 600 volts, 60 Hz, shall be applied for one minute. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord shall meet the requirements of 3.3.3.1.

4.5 Insulation resistance. Insulation resistance shall be in accordance with Method 302 of MIL-STD-202, test condition B. The potential shall be applied between each conductor and the remaining conductors connected together and to the shield, and metallic shells, if present. The cord shall meet the requirements of 3.3.3.2.

4.6 Electrostatic and electromagnetic isolation. The cord shall be tested for electromagnetic and electrostatic isolation to determine compliance with 3.3.3.3 and table I when tested in accordance with figure 3.

4.7 Continuity. One and six tenths (1.6) amperes of direct current shall be passed through each conductor of each cord. The voltage drop across the conductor shall not exceed that specified in 3.3.4.

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RELAY K-1
COIL VOLTAGE. . . . . 6 VDC
COIL CURRENT. . . . . LESS THAN 200m ADC
RELEASE TIME. . . . . LESS THAN 10m Sec
(POTTER AND BRUMFIED TYPE KA11DY OR EQUAL)

FIGURE 4. Monitor circuit for cord flex tester.

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TABLE II. Cable inspections.

Inspection	Paragraph		Group I	Group II
	Require-ments	Test	Qualifica-tion tests	Sample testing per 4.5.1 of the general specification
Tensile Strength	3.2.1.1	4.1	X	X
Temperature	3.3.1	4.2	X	X
Flex Life	3.3.2	4.3	X	X
Dielectric Withstanding Voltage	3.3.3.1	4.4	X	X
Insulation Resistance	3.3.3.2	4.5	X	X
Electrostatic and Electro-magnetic Isolation	3.3.3.3	4.6	X	X
Continuity	3.3.4	4.7	X	X

Custodians:
Army - CR
Air Force - 85

Review activities:
Air Force - 11
Air Force - 99

Preparing activity:
Navy - AS

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